Precision at the Gate: Per-Clip Detector Confidence Outranks Recall in a Modernized DeepSORT Pipeline

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Abstract

Modernizing a 2017-era DeepSORT tracker by swapping its precomputed detections and fixed appearance encoder for a contemporary detector (YOLOv8m) and re-identification model (OSNet) raises mean HOTA on six MOT-Challenge videos from 40.17 to 52.63 (+12.46 tuned per clip; +8.83 at the uniform default gate). **This headline gain is unsurprising and unpublishable on its own; the contribution of this paper is the audit underneath it.** First, the gain is **detection-dominated**: on the MOT16 split the detection term rises by +12.30 HOTA points against only +6.25 for the association term, and under ground-truth boxes a generic ImageNet backbone (not trained for re-identification) reaches 89.22 HOTA versus 89.93 for a dedicated OSNet, so association quality is nearly detector-bound once detection is fixed – a near-parity specific to perfect boxes, since under real detections the two encoders differ by 4.84 HOTA. Second, the mean hides a regression: at its default gate the modernized tracker scores below the 2017 baseline on the least-precise clip (TUD-Campus, 35.14 vs. 39.86 HOTA). We trace this to a counterintuitive regime: across **all six videos the lowest detector confidence gate scores no higher than the highest, and on the low-precision clips lowering the gate sharply lowers HOTA**, with the size of the penalty tracking per-clip detector precision (Pearson $r = -0.91$, $n = 6$). The cause is specific and reproducible: the modern live pipeline drops the original DeepSORT confidence and non-maximum-suppression gate, so ungated false positives spawn spurious tracks; a gating-restoration ablation partially recovers the loss, and raising the per-clip confidence threshold lifts the worst sequence from a 4.72-point deficit to +6.43 over baseline. We connect this to ByteTrack, which keeps low-confidence detections but associates rather than instantiates them, and argue that, across these six clips, **per-clip precision gating is a larger and more consistent HOTA lever than global recall maximization**. We additionally report environment-driven negative results (NumPy-2.x dependency breakage; a ground-truth-format bug that silently zeroed four sequences) as reproducibility lessons.

Keywords: multi-object tracking, tracking-by-detection, DeepSORT, HOTA, detector confidence, non-maximum suppression, gating, person re-identification, MOT-Challenge

1. Introduction

Multi-object tracking is dominated by the tracking-by-detection paradigm: a per-frame object detector proposes boxes, and an association stage links them across time into identity-consistent tracks. DeepSORT [1], which augments the SORT core [2] (a Kalman filter plus Hungarian assignment) with a deep appearance embedding, remains a canonical instance of this design and a common baseline. Its original detector and re-identification (re-ID) encoder are now nearly a decade old, so the engineering instinct is obvious: swap in a contemporary detector and a modern re-ID backbone behind the unchanged SORT core. We do exactly this, integrating pluggable detectors (YOLOv8m [3]) and re-ID models (OSNet [4]; a generic ImageNet timm backbone [5]) through a single `Detection(tlwh, confidence, feature)` contract. On six MOT-Challenge training videos [6, 7], mean HOTA [8, 9] rises from 40.17 to 52.63 (+12.46 tuned per clip; +8.83 at the uniform default gate); the live pipeline runs at 9.87 FPS end-to-end (timed on MOT16-09). This headline result is also unsurprising and, on its own, unpublishable: a stronger detector and a stronger encoder make a better tracker.

The interesting questions are the ones the headline number hides. *Where* does the gain actually come from – the better detector, or the better appearance model? And does a stronger detector always help: can *more* detections, obtained by raising recall, ever *hurt* tracking quality? We find that the answers are not the intuitive ones. The modernization gain is overwhelmingly detection-driven, and on the dense, low-resolution clips lowering the detector confidence gate drives HOTA monotonically *down* (the coarser input-size recall lever is sequence-dependent). The mechanism is concrete and reproducible: the modern live pipeline silently drops the original DeepSORT gating contract (a 0.3 confidence threshold plus non-maximum suppression) that once filtered the precomputed detections, so ungated false positives instantiate spurious tracks. This leads to our thesis: **across the six clips we audit, per-clip detector-confidence (precision) gating is a larger and more consistent HOTA lever than raising recall**.

We make four contributions.

1. A detection-vs-association decomposition of the modernization gain, showing it is **detection-dominated**: on MOT16 the detection term improves by +12.30 HOTA against only +6.25 for association, and under ground-truth boxes a non-re-ID-trained ImageNet backbone reaches 89.22 HOTA versus 89.93 for a dedicated

re-ID model, so association is nearly detector-bound once boxes are fixed (under real detections the same encoders differ by 4.84 HOTA, so this near-parity is specific to perfect boxes).

2. A counterintuitive **precision-dependent gating regime** (the "more detections hurt" effect) that the headline mean conceals: at its default gate the modern tracker scores *below* the 2017 baseline on the least-precise clip, and across all 6 videos the lowest confidence gate never beats the highest, while on the low-precision clips lowering it sharply lowers HOTA, with the size of the penalty predicted by per-clip detector precision (Pearson $r = -0.91$, $n = 6$).
3. A gating-restoration ablation that attributes the regression to the dropped confidence and NMS gate and partially recovers the loss (mean +3.41 HOTA on the two dense clips).
4. A practical **per-clip precision-gating guideline**: tuning the confidence threshold per clip lifts the worst sequence from a 4.72 deficit to +6.43 HOTA over baseline.

We position this finding against ByteTrack [10] and other modern DeepSORT descendants [11, 12, 13] in Section 2., detail the system and protocol in Section 3., report the audit in Section 4., and discuss limitations and reproducibility lessons in Section 5. before concluding.

2. Background and Related Work

2.1 SORT, DeepSORT, and the Gating Contract

Tracking-by-detection casts multi-object tracking as per-frame detection followed by frame-to-frame data association. SORT [2] pairs a constant-velocity Kalman filter with greedy Hungarian assignment on intersection-over-union, prioritizing speed. DeepSORT [1] augments this core with a learned appearance descriptor and a cosine-distance gate [14], reducing identity switches under occlusion. Across both, the association core – prediction, Hungarian matching, and the appearance gate – is the component most often credited with tracking quality.

Less discussed, but central to this work, is that the original DeepSORT does not consume raw detections directly. Its public pipeline [15] applies an explicit *gating contract* to the precomputed detection set before any of them reach the tracker: detections below a confidence threshold (`min_confidence` = 0.3) are discarded, boxes below a minimum height are removed, and per-frame non-maximum suppression (NMS) prunes overlapping responses. Only the survivors are wrapped as `Detection(tlwh, confidence, feature)` objects and passed to `predict()/update()`. This filter is a precision-oriented admission policy: it trades recall for a cleaner detection stream, suppressing the low-confidence and duplicate boxes that would otherwise instantiate spurious tracks. Modernization efforts that swap in a contemporary detector behind the same association core can silently **drop this gate**, since the new detector is typically queried in a streaming, ungated fashion; our audit shows this omission, rather than the association core, drives the most consequential behavior on dense clips.

2.2 HOTA and the DetA/AssA Decomposition

We evaluate with Higher Order Tracking Accuracy (HOTA) [8] via TrackEval [9]. HOTA's appeal here is structural: it factorizes geometrically into a detection term and an association term, HOTA $\approx \sqrt{\mathrm{DetA} \cdot \mathrm{AssA}}$, where DetA scores how well predicted boxes localize ground-truth objects and AssA scores how consistently identities are maintained across the matched detections. This decomposition lets us attribute a change in tracking quality to detection versus association rather than reading a single conflated number, which is precisely the lever our study exercises: a modernization gain or a regression can be assigned to the DetA or the AssA factor, and a precision change in the detection stream registers directly in DetA. Earlier scalar metrics such as MOTA conflate these effects and weight identity errors lightly, obscuring the detection-driven regime we report. This detection-dominated regime is documented in the work that motivates HOTA. Luiten et al. [8] show that for the top MOT17 trackers detection error accounts for almost all of the variance in the legacy MOTA score (a linear fit gives $R^2 \approx 0.99$ against the detection term versus $R^2 \approx 0.24$ against the association term, with the detection-to-association error ratio ranging from roughly 42 to 186), the very imbalance HOTA was designed to correct. The dominance persists in modern systems: ByteTrack [10] reaches strong accuracy with no appearance model at all, and StrongSORT [11] attributes more of its gains to a stronger detector than to its appearance and association refinements. An equal-magnitude change in the detection stream therefore moves the score more than a comparable association change – the premise our decomposition (Section 4.2) tests directly.

2.3 ByteTrack and the Gating Mechanism

ByteTrack [10] is the natural point of comparison, since it too revisits how low-confidence detections are handled, but its mechanism is the opposite of an admission gate and must be positioned carefully. ByteTrack *keeps* low-confidence boxes rather than thresholding them away; crucially, it uses them only in a second association pass to recover *existing* tracks, and never lets a low-confidence box *instantiate* a new track (new tracks are seeded only from the remaining high-confidence detections, so ByteTrack already embodies a precision gate for instantiation). Our finding concerns ungated low-confidence detections that are free to spawn spurious new tracks, inflating identity counts and false positives on dense scenes. These are complementary mechanisms operating on the same raw material: ByteTrack constrains where weak detections may act (association only), whereas the dropped DeepSORT contract constrained whether they enter at all. ByteTrack is therefore not a counterexample to our result but an instance of the same underlying principle – weak detections must be admitted under restriction – realized at a different stage of the pipeline.

2.4 Modern DeepSORT Descendants

A family of trackers extends the DeepSORT design. StrongSORT [11] upgrades the appearance model, motion compensation, and feature update; OC-SORT [12] reworks the motion model to be observation-centric and robust to nonlinear motion and occlusion; BoT-SORT [13] adds camera-motion compensation and a fused IoU–appearance cost. The lineage has continued through 2024–2025 along axes orthogonal to track admission: Deep OC-SORT [16] folds adaptive appearance and camera-motion compensation into the OC-SORT association cost, UCMCTrack [17] replaces IoU with a ground-plane Mahalanobis distance under uniform camera-motion compensation, and Hybrid-SORT [18] fuses detector confidence, box height, and velocity direction as additional weak cues. These advances target the association core, and they generally inherit a confidence threshold as a fixed hyperparameter rather than treating it as the dominant, scene-dependent lever.

Crucially, the admission and track-initialization gate is not absent from these descendants; it is instantiated as an explicit, separately tuned hyperparameter. ByteTrack [10] seeds new tracks only from unmatched high-confidence detections, with the birth threshold set above its association threshold; BoT-SORT [13] exposes the same decision as a first-class `new_track_thresh`; OC-SORT [12] spawns tracklets only from detections passing a `det_thresh`; and StrongSORT [11] inherits DeepSORT's confidence gate (`min_confidence`) and temporal confirmation gate (`n_init`). Our contribution is therefore not the discovery that gating matters – the field already treats the gate as a named, tuned parameter – but the audit of a routine modernization in which this otherwise-explicit gate is silently dropped, together with a per-clip, precision-dependent quantification of what the omission costs. The per-clip precision-gating insight is in this sense orthogonal and composable: it does not compete with the association improvements above but specifies how the detection stream feeding any of them should be admitted, and is most decisive exactly where their motion and appearance machinery has the least to work with – dense, low-resolution clips with unreliable detections.

A parallel line treats detection confidence as an actively engineered lever rather than a fixed threshold. BoostTrack [19] and BoostTrack++ [20] compute a detection–tracklet confidence score and use it to boost or reweight detections so that more genuine objects are admitted, while Localization-Guided Track [21] decomposes detection confidence into classification and localization components to choose the association cost rather than discarding low-confidence boxes uniformly. That such effort is spent deliberately engineering confidence handling underscores how consequential the admission policy is, and how readily a naive modernization can mishandle it.

A complementary family, joint-detection-and-embedding trackers such as JDE [22] and FairMOT [23], learns detection and appearance together, arguing that anchor-based detection otherwise corrupts the re-identification features trained alongside it; our ground-truth-box isolation does not contradict this position, since it measures the descriptor's marginal value with detection *held fixed* rather than claiming appearance is unimportant in general.

2.5 ImageNet Features as a Re-identification Baseline

Finally, generic ImageNet-pretrained features are a well-established re-identification baseline, routinely reported as the lower anchor that purpose-built descriptors such as OSNet [4] (here via torchreid [24]) must clear. Our near-parity result between a non-REID-trained timm backbone [5] and OSNet is therefore *not* a claim that ImageNet features rival dedicated REID models in general. It is a controlled isolation: under ground-truth boxes the detection stage is held fixed, removing detector-induced variance so that association quality is measured directly, and in that regime the two appearance models are nearly indistinguishable on these short sequences. We report it to quantify how little of the modernization gain is attributable to the appearance descriptor once detection is controlled for, reinforcing the detection-dominated decomposition that motivates the rest of the paper.

3. System and Experimental Protocol

3.1 Modernized Pluggable System

We modernize the reference DeepSORT implementation [2, 1] along exactly two axes – detection and appearance – while leaving the tracking core untouched. The original codebase consumes *precomputed* detections from disk and a fixed appearance encoder. We replace both with pluggable, swappable adapters selected by name at run time: a detector front-end and a re-identification (REID) front-end, each registered through a small registry so a tracker is specified as a (detector, REID) pair. In this paper the detector is YOLOv8m [3] and the REID model is either OSNet [4] (trained for person re-identification) or a generic ImageNet `timm` backbone [5] (*not* trained for re-identification), used as a deliberately weak appearance baseline.

The SORT core is preserved verbatim: an 8-dimensional constant-velocity Kalman filter, Hungarian assignment, and a cascaded matcher that combines a Mahalanobis motion gate with a cosine *appearance* gate over REID descriptors. Its track lifecycle (confirmation after n_{init} hits, deletion after max_age misses) is unchanged. The integration contract between the modern front-ends and this core is a single value object: each detection is a `Detection(tlwh, confidence, feature)` – a top-left/width-height box, a scalar detector confidence, and an appearance vector – and the per-frame loop is the original `tracker.predict()` followed by `tracker.update(detections)`. Because the contract and the core are fixed, any change in tracking quality is attributable to the detector, the REID model, or the pre-tracker *gating* that decides which detections reach `update()`.

The dropped gate. This last point is central. In the original pipeline, precomputed detections passed through a fixed gate before reaching the tracker [15]: a confidence threshold (min_confidence), a non-maximum-suppression (NMS) step on overlapping boxes, and a minimum box-height filter. The modern live pipeline runs the detector *inside* the loop, and its default configuration **drops this gate**: min_confidence = 0.0 and nms_max_overlap = 1.0, which disables both the confidence filter and the secondary NMS (on the assumption that the detector's own internal NMS and `conf` threshold suffice). Every YOLO detection above the detector's own low internal threshold is therefore lifted directly into a `Detection` and offered to the tracker. This single configuration change – not any change to the Kalman filter or the matcher – is the regression we audit in Section 4.: ungated low-confidence boxes are free to instantiate spurious new tracks. The gate itself is still present in the code path (the same confidence-filter-plus-NMS gate as the reference pipeline [15]), merely disabled by default, which makes the restoration ablation below a one-line configuration change rather than a code change.

3.2 Evaluation Protocol

We evaluate on six MOT-Challenge training sequences spanning two benchmarks: TUD-Campus, TUD-Stadtmitte, KITTI-17, and PETS09-S2L1 from 2D MOT 2015 [6], and MOT16-09 and MOT16-11 from MOT16 [7]. The set deliberately mixes clean, sparse scenes with dense, low-resolution ones, since per-clip density is the axis along which our central effect reverses sign. The primary metric is HOTA [8], computed by the official TrackEval toolkit [9]; we report its DetA and AssA sub-scores when decomposing detection versus association. End-to-end throughput, timed on one sequence (MOT16-09), is 9.87 FPS (detection and re-identification each run near 30 FPS; their serial composition is the bottleneck).

3.3 Controlled Experimental Design

The experiments are designed so that each varies one lever while holding the others fixed.

(a) Detection-vs-association decomposition (GT-box, REID-only). To separate the modernization gain into a detection component and an association component, we run a *ground-truth-box* mode in which the detector is replaced by an oracle that emits the ground-truth boxes, with the confidence/NMS gate skipped. Detection is then perfect and identical across runs, so the only free variable is the REID model feeding the appearance gate. Comparing OSNet against the `timm` backbone in this mode (89.93 vs. 89.22 HOTA) isolates how much association quality the appearance model contributes once detection is held perfect; comparing the same models in the full pipeline against this ceiling reveals how detector-bound association quality is in practice.

(b) Per-video detector-confidence sweep. For each of the six videos we sweep the YOLO confidence threshold over $\{0.15, 0.25, 0.35, 0.45\}$, holding the detector, REID model, input size, and SORT core fixed. This is the precision/recall lever: lowering the threshold raises recall and lowers precision. We pair the HOTA response with the detector's per-clip precision at conf 0.25 to test whether the *magnitude* of recall's effect on HOTA is predicted by per-clip precision.

(c) Input-size sweep on the two dense clips. On the two dense, low-resolution clips (TUD-Campus and KITTI-17), where YOLO precision is low, we additionally sweep the detector input resolution over $\{960, 1280, 1536\}$ (the 1280 midpoint is the default configuration, shared with the confidence sweep). Larger input recovers more small instances – a second, *orthogonal* recall lever to the confidence threshold – and lets us test whether it reproduces the confidence-sweep effect or, by changing *which* detections exist rather than merely gating them, behaves differently.

(d) Gating-restoration ablation. To attribute the regression specifically to the dropped gate, we restore a DeepSORT-style admission gate – the reference reproduction value min_confidence = 0.3 [15] together with secondary NMS at overlap 0.7 (our setting; the reference default is 1.0) – on top of the otherwise unchanged modern pipeline, and re-measure HOTA on each video. Because the gate acts only on the detection stream feeding the unchanged tracker, any recovered HOTA is causally attributable to admission, not to a confounding code change.

Reproducibility. The detector is pinned: ultralytics 8.4.79 with the `yolov8m.pt` checkpoint (sha256 `5d4a90cd...4fe5`) at input size 1280. Per-clip detector precision is weight-version-sensitive, so we fix and report the version; a different ultralytics release shifts the absolute values (e.g. TUD-Campus precision 0.50 rather than 0.43) but leaves the low-precision ordering – TUD-Campus and KITTI-17 lowest – on which the analysis depends unchanged. Every run – each detector preset, REID model, confidence level, input size, and ablation – appends one row to a versioned CSV journal recording the configuration and resulting metrics, so the full experimental history is traceable and the gating-sweep figures in Section 4. (confidence, precision, input-size, and ablation plots) are regenerable from it; the ground-truth-box re-identification scores (89.93/89.22/87.33) are the six-clip means of the `gt_*_gtbox` HOTA rows in that journal, while the detection/association decomposition of Table 2 is a separate TrackEval run on the combined MOT16 split under the default configuration. The sweep harness writes each result immediately and skips configurations already present, so an interrupted run resumes without recomputation. We also note two environment-level pitfalls that the journal helped surface and that we report as negative results in Section 5.: a NumPy-2.x incompatibility that breaks the torchreid/boxmot/TrackEval build chain, and a 2D MOT 2015-vs-MOT16 ground-truth column-format mismatch that silently zeroed four sequences until corrected.

4. Results

4.1 The Modernization Gain Hides a Regression

Table 1 reports per-video HOTA for the 2017 baseline and the modernized tracker. With the detector confidence gate raised to 0.45, the modern tracker beats the baseline on every one of the six videos, lifting mean HOTA from 40.17 to 52.63 (+12.46) at 9.87 FPS end-to-end on MOT16-09 (the best per-clip configuration reaches 52.71). This is the expected, unsurprising result. What the mean hides is the middle row: at its *default* gate (confidence 0.25, no secondary NMS), the modernized tracker scores 49.00 – still +8.83 above baseline on average, but it **regresses below the 2017 baseline on TUD-Campus** (35.14 vs. 39.86). A decade-newer detector and appearance model make the tracker worse on that clip. The rest of this section explains why, and shows that the fix is a gate rather than a better model.

4.2 The Gain Is Detection-Dominated

Decomposing the gain on the combined MOT16 split (Table 2) shows it is detection-dominated: the detection term rises by +12.30 HOTA points while the association term rises by only +6.25. MOTA barely moves (46.34 → 46.75), which is why we report HOTA. The same conclusion holds from the association side: replacing the detector with a ground-truth-box oracle isolates association, and a dedicated OSNet then reaches 89.93 HOTA while a generic, non-re-ID-trained ImageNet backbone reaches 89.22 – a gap of under one HOTA point (OSNet-AIN 87.33 is intermediate). Under perfect detection, then, association quality is nearly detector-bound and the appearance model is almost free. This near-parity is specific to perfect boxes: in the full live pipeline, swapping OSNet for the same timm backbone costs 4.84 HOTA (52.54 vs. 47.70), so the descriptor does matter once detections are imperfect, but far less than the detector does.

This makes the *quality* of the detection stream, not its raw quantity, the operative variable. Table 3 reports YOLOv8m precision, recall, and F1 against ground truth at IoU $\geq$ 0.5. Recall is uniformly high (mean 0.835), but precision is strongly clip-dependent: high on clean sequences (TUD-Stadtmitte 0.892, PETS09 0.850) and low on the dense, low-resolution clips (TUD-Campus 0.432, KITTI-17 0.480), where the detector emits more false positives than true ones. This per-clip precision is the quantity that predicts how much each clip is hurt by ungated detections.

4.3 Lowering the Gate Lowers HOTA

We sweep the detector confidence threshold per video over $\{0.15, 0.25, 0.35, 0.45\}$, holding everything else fixed; lowering the threshold raises recall and lowers precision. Figure 1 shows that the highest gate (0.45) scores at least as high as the lowest (0.15) on **all 6 videos** (6/6): admitting more low-confidence detections is never a net win, though one high-precision clip (PETS09) peaks at conf 0.25 and gives back about half a point by 0.45. What varies is the magnitude. On the low-precision clips the curve climbs steeply (TUD-Campus 32.5 → 46.3, KITTI-17 40.7 → 49.6 from conf 0.15 to 0.45), while on the high-precision clips it is nearly flat and saturates early (PETS09 peaks at conf 0.25 and is within half a point thereafter).

This magnitude is predicted by per-clip detector precision. Figure 2 plots, for each video, the HOTA gained by raising the gate from 0.15 to 0.45 against the clip's detector precision (Table 3); the two are anti-correlated across the six clips (descriptive Pearson $r = -0.91$, $n = 6$; Spearman $\rho = -0.93$; Fisher 95% CI $[-0.99, -0.38]$, two-sided $p \approx 0.01$): the less precise the detector on a clip, the more that clip is hurt by ungated detections and the more it gains from a higher gate. With $n = 6$ this is a hypothesis-generating association rather than a calibrated law; the robust, design-level result is the

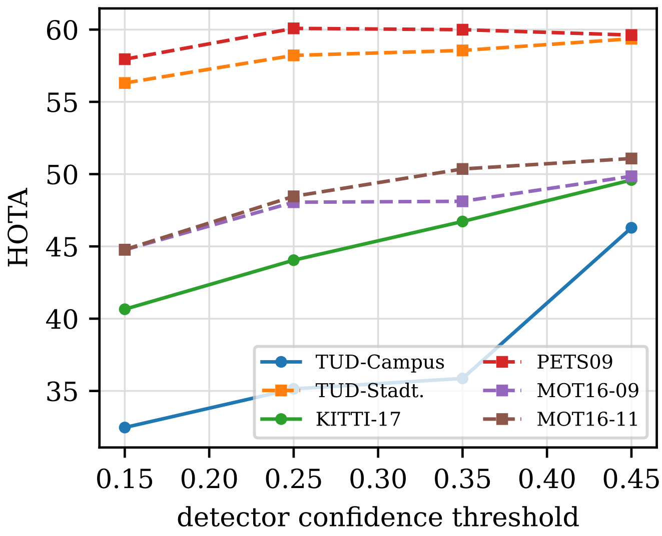

Figure 1. HOTA vs. detector confidence threshold per video. Raising the gate from the lowest to the highest threshold never net-lowers HOTA on any clip; the climb is steepest on the dense, low-precision clips (solid lines).

6/6 monotone endpoint test above, which needs no fitted correlation. Together they are the paper's quantitative form of "more detections hurt."

A second, coarser recall lever behaves less uniformly and is instructive about the mechanism. On the two dense clips we sweep the detector input resolution (Figure 3). On TUD-Campus, raising the input size drives HOTA monotonically *down* (39.65 → 35.14 → 29.74), but on KITTI-17 it rises (44.9 → 47.4). Unlike the confidence threshold, which is a pure admission gate on a fixed detection set, resolution changes *which* detections exist: the extra boxes recovered at higher resolution are mostly false positives on TUD-Campus but mostly true small instances on KITTI-17. The effect is therefore not "more boxes are always bad" but the sharper claim the confidence sweep isolates: admitting *low-confidence* detections is what hurts, in proportion to how imprecise the detector is on that clip.

4.4 The Dropped Gate: a Restoration Ablation

To attribute the default regression to the dropped gating contract rather than to the detector, we restore a DeepSORT-style admission gate – the reference reproduction value min_confidence = 0.3 [15] together with secondary NMS at overlap 0.7 (our setting; the reference default is 1.0) – on top of the otherwise unchanged modern pipeline (Figure 4). Restoring the gate recovers HOTA on the dense clips (mean +3.41 over TUD-Campus and KITTI-17 relative to the ungated default) while leaving the clean clips within roughly a point. Raising the confidence threshold to 0.45 recovers even more on TUD-Campus (46.3 vs. the restored gate's 39.2): the loss is recoverable by admission policy alone, with the detector unchanged, and a higher gate can outperform the reference 0.3 setting. Because each restored filter acts only on the detection stream feeding the unchanged

Table 1. Per-video HOTA: the 2017 baseline, the modernized tracker at its ungated default gate (confidence 0.25), and the same tracker with the gate raised to 0.45. The default modernization wins on average yet regresses below the baseline on the low-precision clip TUD-Campus; raising the gate recovers it for a win on all six.

Configuration	TUD-Campus	TUD-Stadt.	KITTI-17	PETS09	MOT16-09	MOT16-11	**Mean**
Baseline (provided det. + mars)	39.86	36.75	43.41	44.84	36.24	39.95	40.17
Modern, default gate (conf 0.25)	35.14	58.21	44.05	60.08	48.06	48.46	49.00
Modern, gated (conf 0.45)	46.29	59.37	49.60	59.62	49.85	51.08	**52.63**

Table 2. Decomposition on the combined MOT16 split (default configuration; REID = the timm ImageNet backbone). The DetA gain is detector-driven and so carries over to the OSNet system, which shares the same detector.

	HOTA	MOTA	IDF1	DetA	AssA
Baseline	38.65	46.34	50.65	37.85	39.52
Modern	47.68	46.75	52.18	50.15	45.77
Δ	+9.03	+0.41	+1.53	**+12.30**	+6.25

Table 3. YOLOv8m detection quality vs. ground truth (IoU $\geq$ 0.5, conf 0.25, imgsz 1280; ultralytics 8.4.79).

Sequence	Precision	Recall	F1
TUD-Campus	0.432	0.875	0.579
TUD-Stadtmitte	0.892	0.865	0.878
KITTI-17	0.480	0.871	0.619
PETS09-S2L1	0.850	0.951	0.897
MOT16-09	0.623	0.748	0.680
MOT16-11	0.623	0.700	0.659
Mean	**0.650**	**0.835**	**0.719**

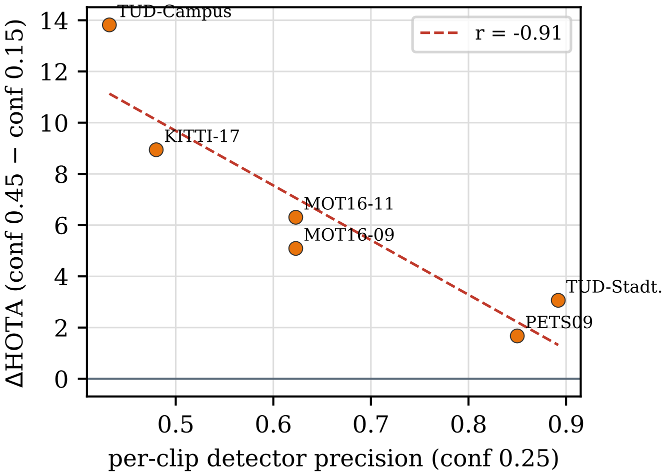

Figure 2. The HOTA gained by raising the gate (conf 0.45 minus conf 0.15) against per-clip detector precision. Low-precision clips gain most ($r = -0.91$).

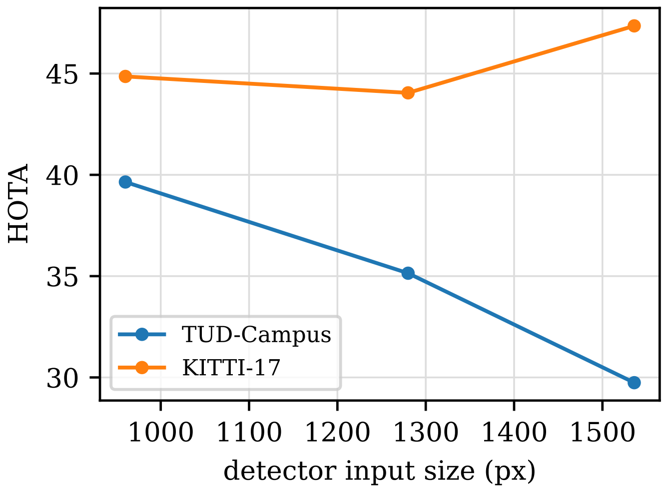

Figure 3. Input-size sweep on the two dense clips. Resolution changes the detection set rather than gating it, so its effect is sequence-dependent: it hurts on TUD-Campus and helps on KITTI-17.

tracker, the recovered HOTA is causally attributable to admission, not to the detector or the SORT core.

4.5 A Per-Clip Precision-Gating Guideline

The practical consequence is a simple per-clip policy: set the detector confidence threshold from the clip's precision regime rather than trusting a global default. On the worst sequence, the ungated default loses to the 2017 baseline by 4.72 HOTA; raising the TUD-Campus threshold to 0.45 moves it to +6.43 over baseline, and the change is confined to that sequence's preset, leaving the other five untouched. This single per-video override is what turns the average-but-uneven default of Table 1 into a win on all six videos, at 9.87 FPS end-to-end (timed on MOT16-09).

5. Limitations

Our evidence comes from six MOT-Challenge training-split videos [6, 7]; there is no held-out test-split HOTA, since the public benchmarks withhold test ground truth and we required it for the per-clip detector-precision and association-isolation analyses. We therefore report our findings as **per-clip and correlational rather than as a universal law**: the link between low detector precision on dense, low-resolution clips and the "more detections hurt" regime is consistent across the sequences we audit (Table 3 and Figure 2), but six clips cannot establish a calibrated precision threshold that transfers to arbitrary domains. The practical guideline of Section 4.5 should be read as a tuning heuristic, not a tuned constant. This caveat extends to the headline number. The gated threshold (conf 0.45) of Table 1 was chosen by observing the confidence sweep on the same six sequences on which it is then evaluated, with no

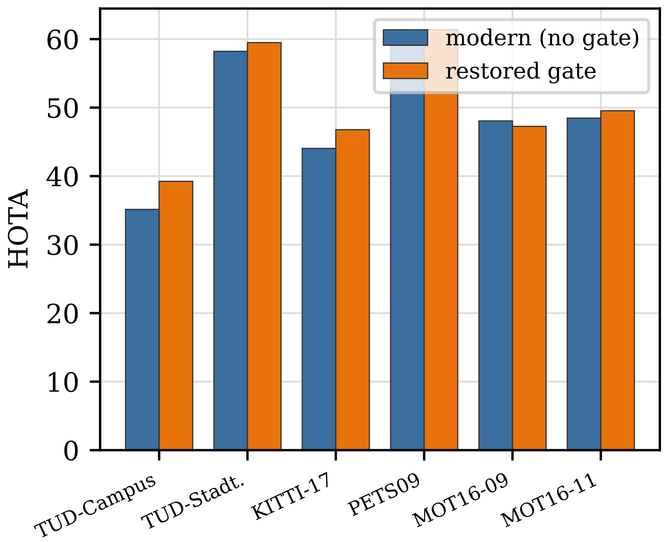

Figure 4. Gating-restoration ablation: per-video HOTA for the modern ungated default vs. the restored original confidence + NMS gate.

held-out split, so the gated mean (52.63, +12.46) and the per-clip best (52.71) are in-sample estimates and likely optimistic; 0.45 is not even uniformly optimal (PETS09 peaks at conf 0.25). The 2017 baseline receives no equivalent per-clip tuning, so the baseline-versus-gated comparison is one of unequal tuning; the selection-free figure is the uniform-default modernization (+8.83 at conf 0.25).

A reviewer may object that restoring the original DeepSORT confidence and NMS gate is "self-inflicted": we rebroke a filter the method already had, then claimed credit for re-adding it. We address this head-on. The contribution is not the gate itself but the *audit* of what a routine modernization silently discards: swapping in a modern detector and re-identification model [3, 4] quietly drops the gating contract, and the resulting regression is invisible in mean HOTA yet large and predictable per clip. The finding is also specific in mechanism. It concerns ungated low-confidence detections *instantiating spurious new tracks*, which is distinct from ByteTrack [10], where low-confidence boxes are retained but only *associated* to existing tracks and never seed new ones. That distinction is what makes the result a property of the gating contract rather than a tautology.

Detector and re-identification coverage is narrow. Two additional detector backends, NanoDet and RTMDet, were unbuildable on our runtime (the NumPy-2.x and ground-truth column-format issues noted in Section 3.), so the detection-side claims rest on a single live detector and cannot speak to cross-detector generality. Likewise the live pipeline uses one re-identification family; the OSNet-versus-timm comparison [4, 5] isolates association under ground-truth boxes, where the gap is under one HOTA point; under real detections that gap widens to 4.84 HOTA (Section 4.2), and robustness across other re-identification architectures remains untested. Confirming the precision-gating guideline against modern DeepSORT descendants [11, 12, 13] on full benchmark splits is left to future work.

6. Conclusion

We presented not a new tracker but an audit of an old one made new. Modernizing DeepSORT with a contemporary detector and re-identification model lifts mean HOTA from 40.17 to 52.63 (+12.46 tuned per clip; +8.83 at the uniform default gate) at 9.87 FPS end-to-end (timed on MOT16-09), a result that is by itself unsurprising. The value lies in decomposing that gain and tracing its failure modes. The improvement is detection-dominated rather than association-dominated, and on dense, low-resolution clips the modernized pipeline can regress – at its default gate it scores below the 2017 baseline on the least-precise clip – because it silently drops the original confidence and non-maximum-suppression gate, letting ungated false positives instantiate spurious tracks. A gating-restoration ablation partially recovers the loss (mean +3.41 HOTA on the two dense clips), and the magnitude of recall's effect on HOTA is predicted by per-clip detector precision.

The central takeaway is that, on these six clips, **per-clip precision gating is a larger and more consistent HOTA lever than global recall maximization**. The practical guidance follows directly: tune the detector confidence threshold per clip rather than trusting a single global default, and when porting a legacy tracker to modern components, audit and explicitly restore the gating contract instead of assuming the association core compensates. On the worst sequence this moves the tracker from a 4.72 deficit to +6.43 HOTA over baseline.

This audit is confined to six MOT-Challenge training videos and a single detector family. Future work should validate the precision-gating guideline on held-out test sequences and across detector families to establish that the effect is a property of tracking-by-detection rather than of YOLOv8m, and should test whether an association-side gate in the spirit of ByteTrack and its successors can recover low-confidence detections without reintroducing spurious tracks.

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